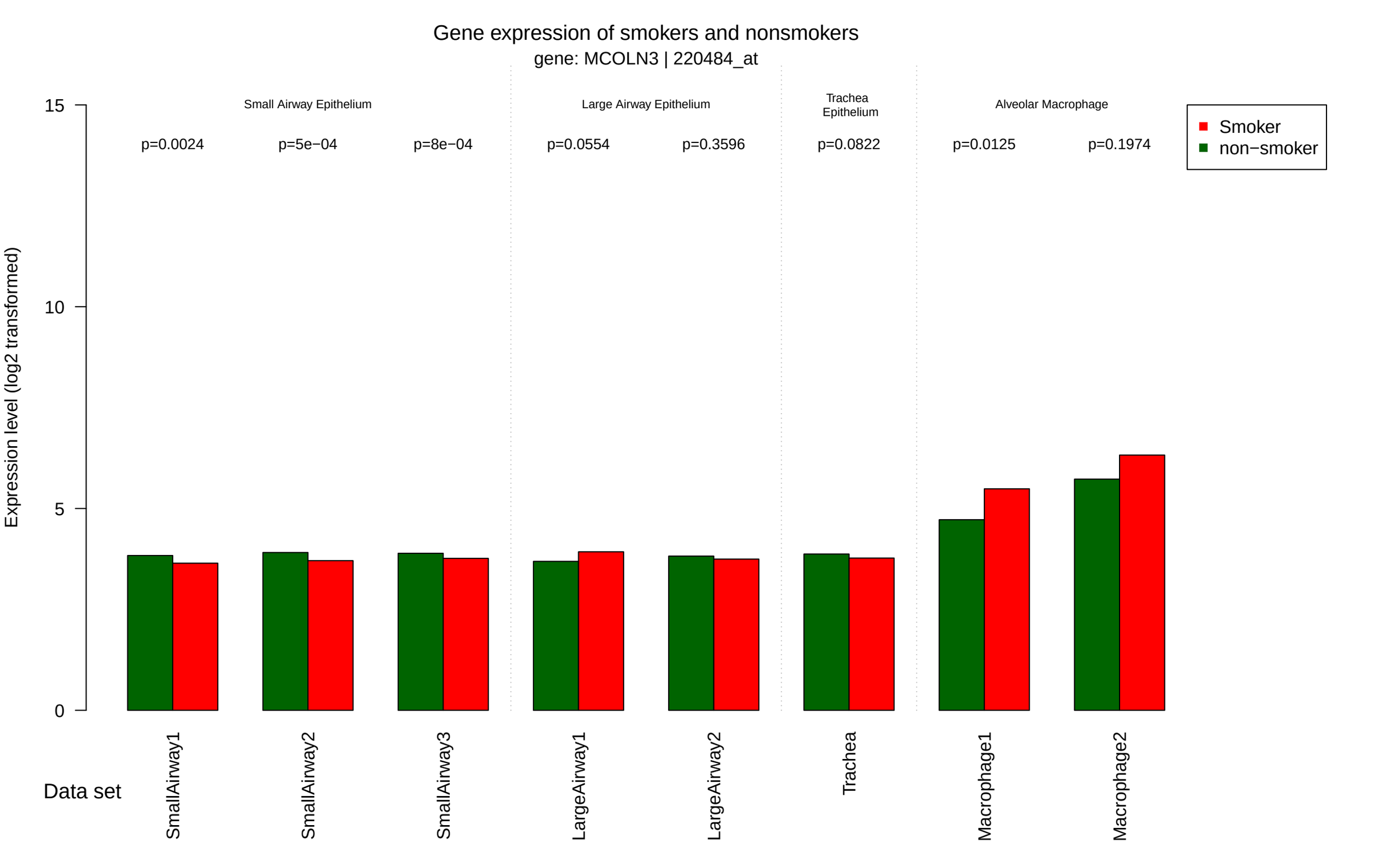
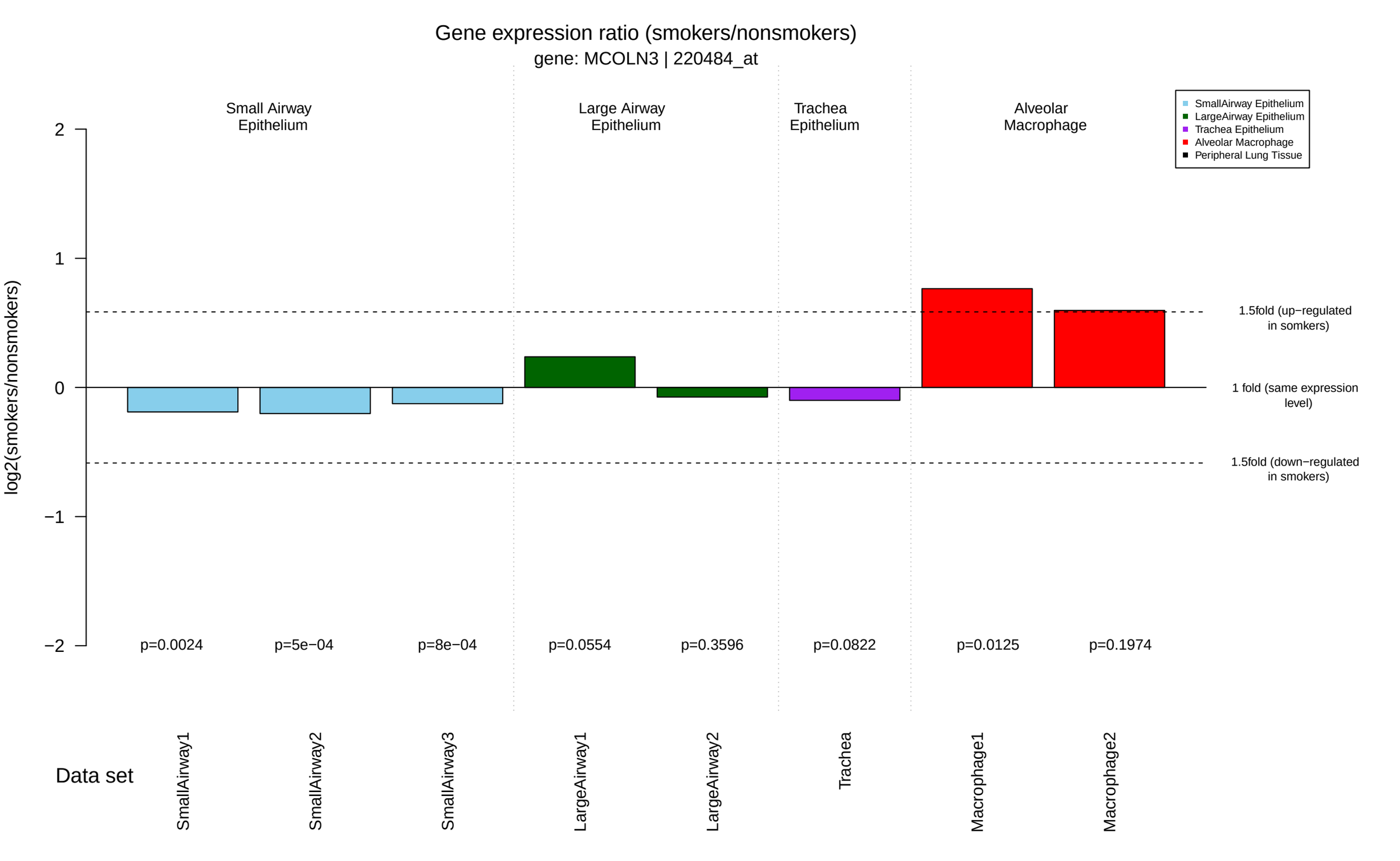
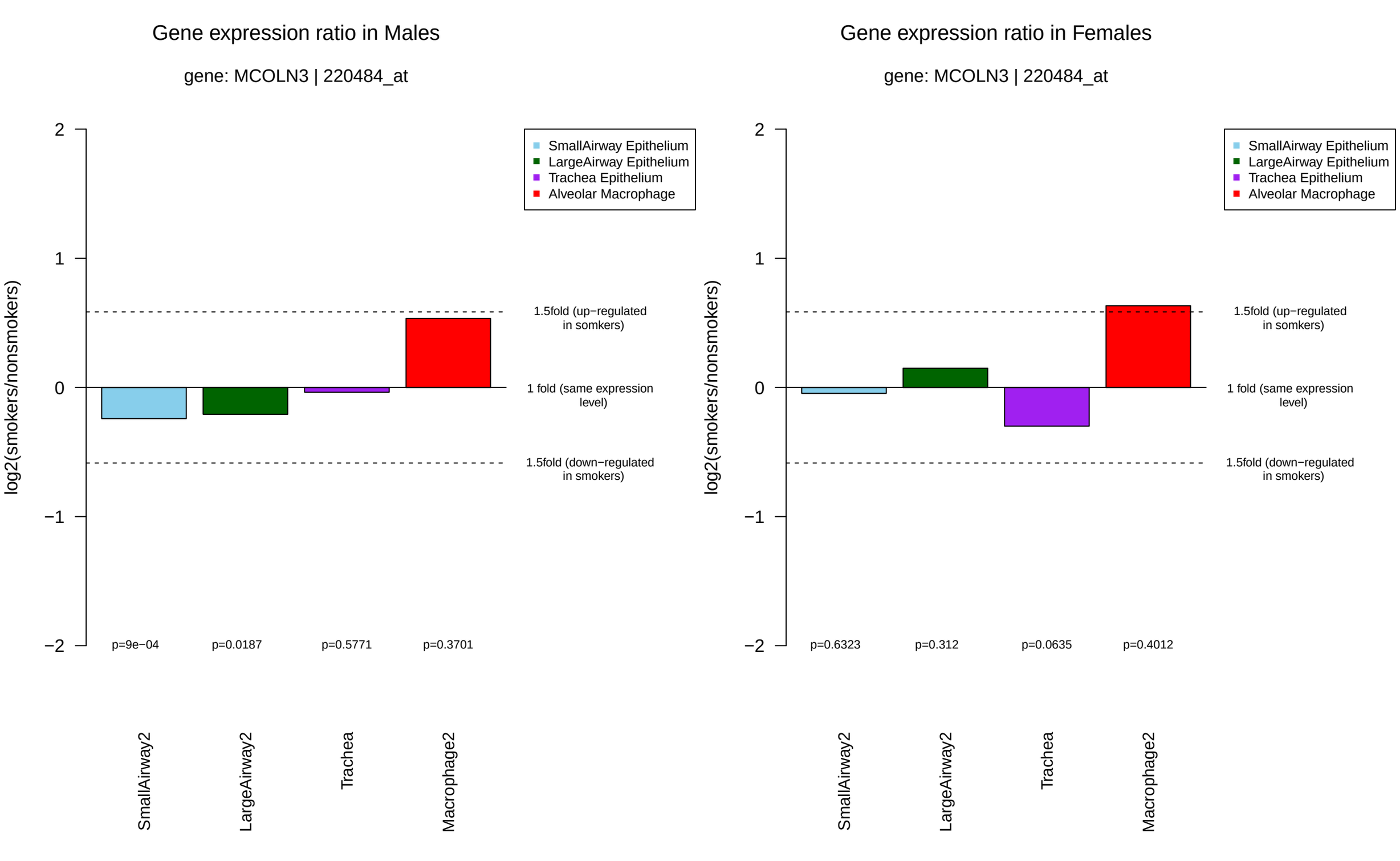
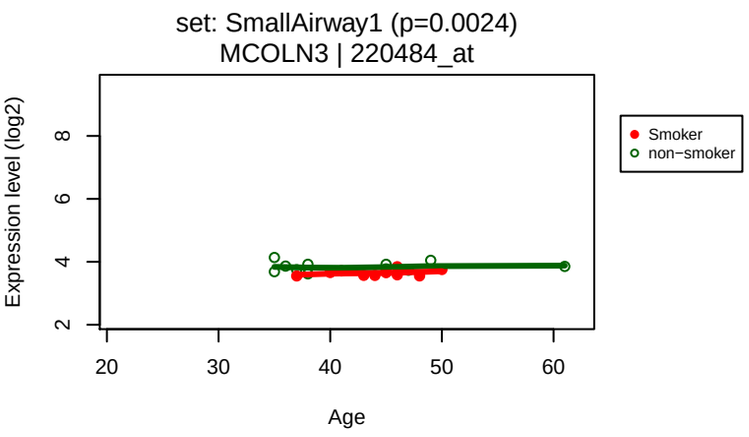
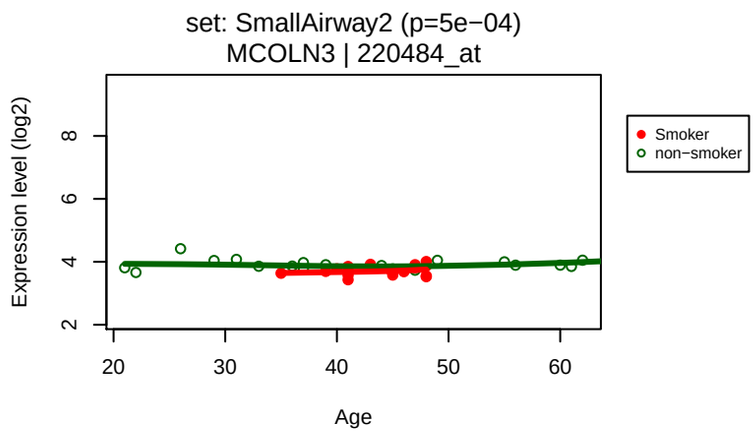
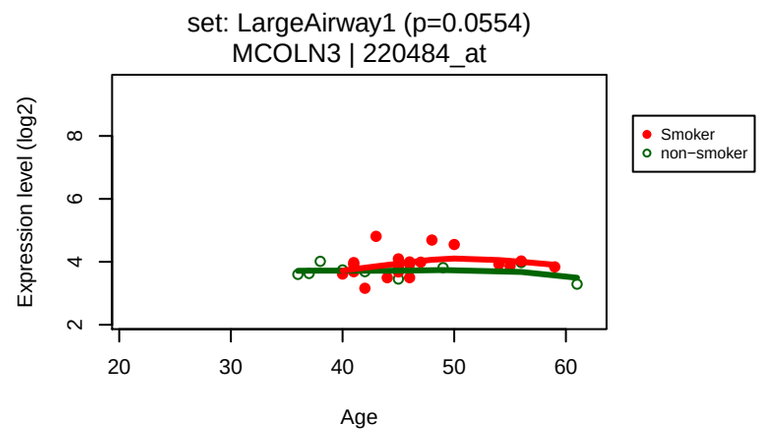

Gene expression of smokers and nonsmokers
gene: MCOLN3 | 220484_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0024
p=5e−04
p=8e−04
p=0.0554
p=0.3596
p=0.0822
p=0.0125
p=0.1974
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

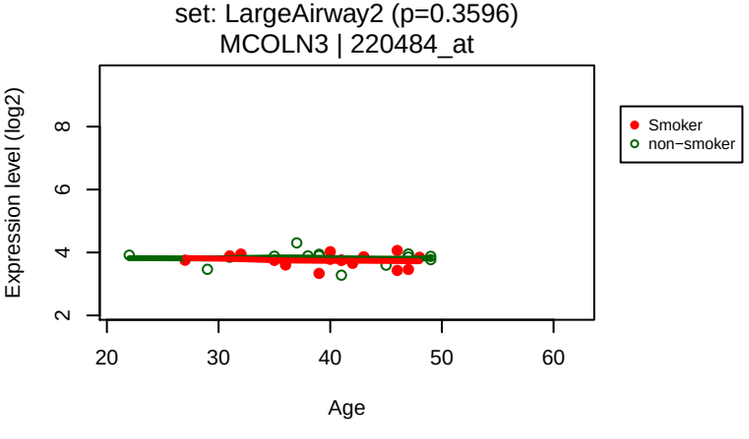

Gene expression ratio (smokers/nonsmokers)
gene: MCOLN3 | 220484_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0024
p=5e−04
p=8e−04
p=0.0554
p=0.3596
p=0.0822
p=0.0125
p=0.1974
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

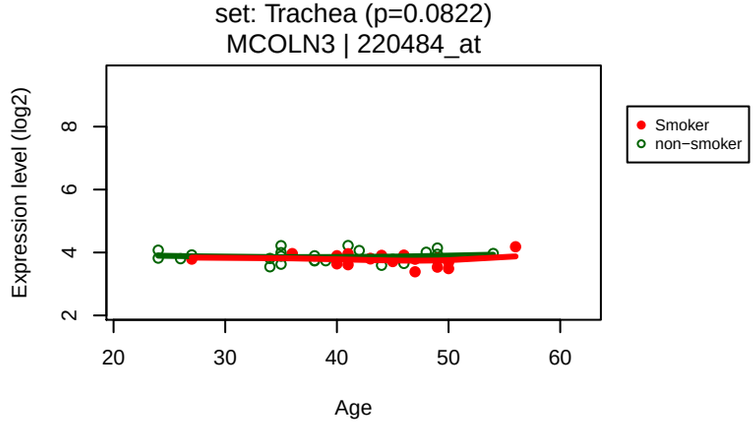

Gene expression ratio in Males
gene: MCOLN3 | 220484_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=9e−04
p=0.0187
p=0.5771
p=0.3701
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MCOLN3 | 220484_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6323
p=0.312
p=0.0635
p=0.4012
SmallAirway2
LargeAirway2
Trachea
Macrophage2

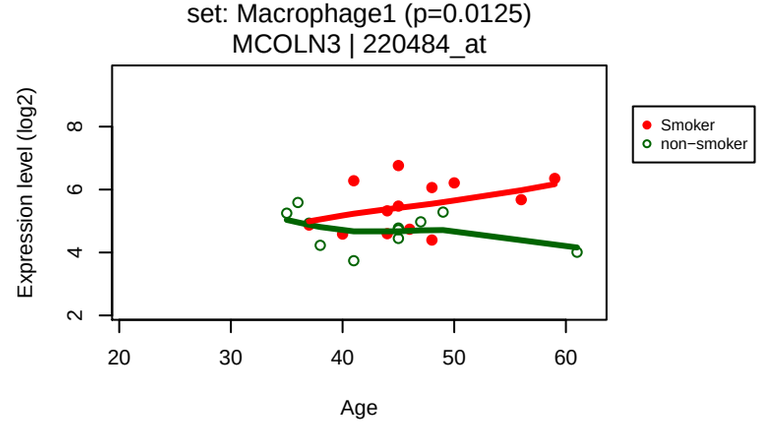

set: SmallAirway1 (p=0.0024)
MCOLN3 | 220484_at
Expression level (log2)
Age
Smoker
non−smoker

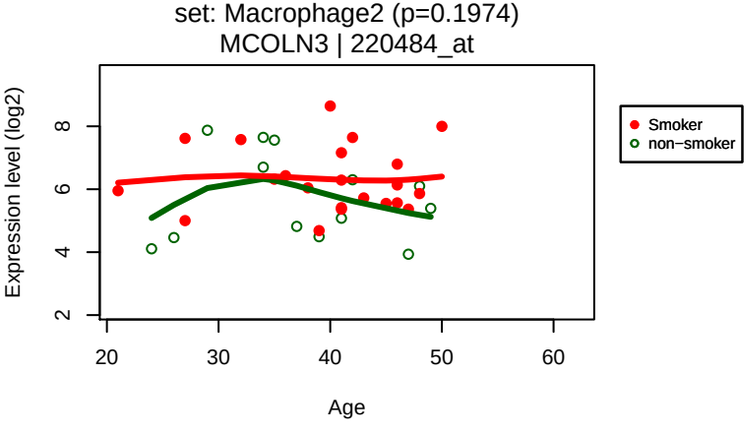

set: SmallAirway2 (p=5e−04)
MCOLN3 | 220484_at
Expression level (log2)
Age
Smoker
non−smoker

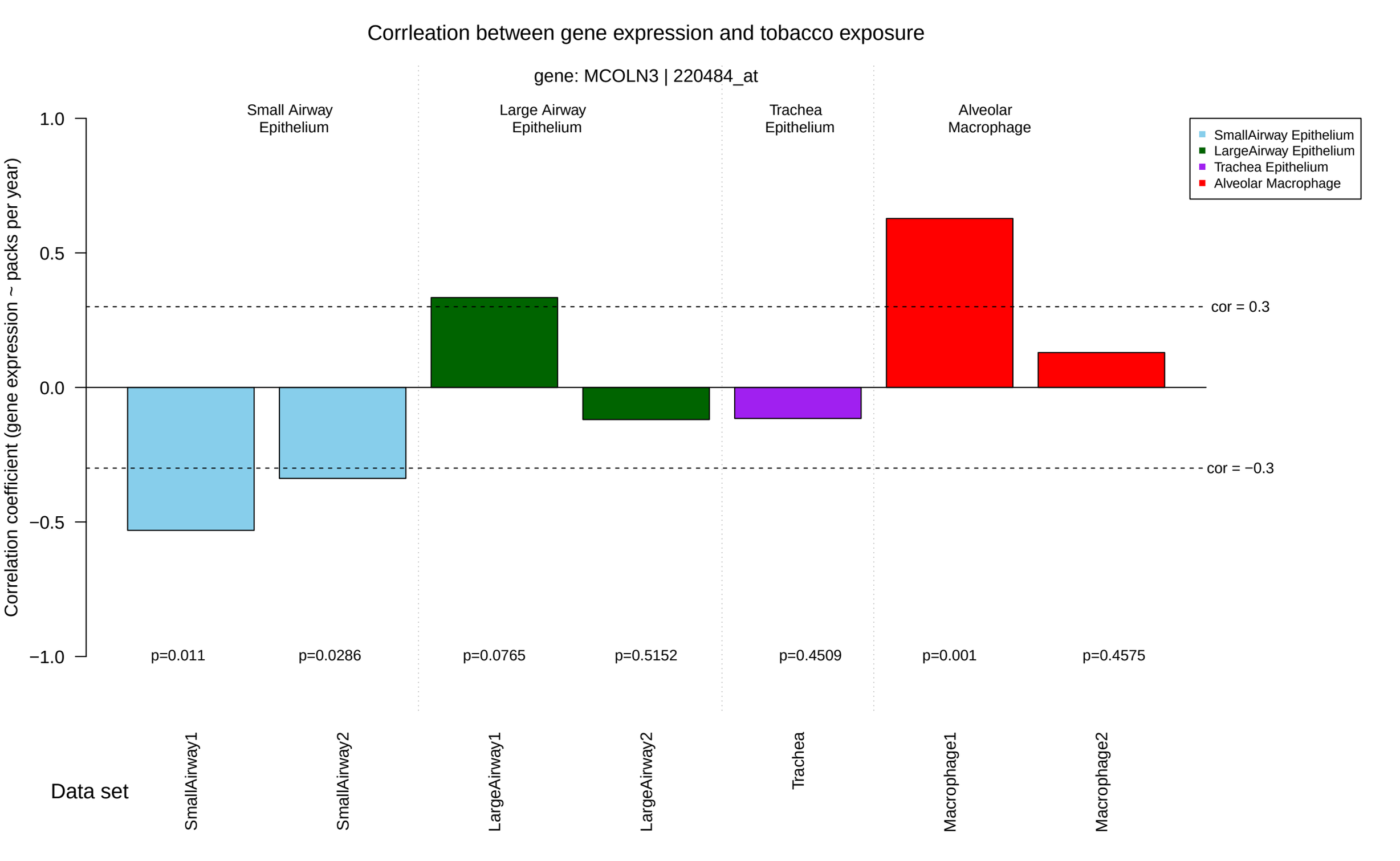

set: LargeAirway1 (p=0.0554)
MCOLN3 | 220484_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3596)
MCOLN3 | 220484_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0822)
MCOLN3 | 220484_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0125)
MCOLN3 | 220484_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1974)
MCOLN3 | 220484_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MCOLN3 | 220484_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.011
p=0.0286
p=0.0765
p=0.5152
p=0.4509
p=0.001
p=0.4575
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2